Pratikshya Regmi

Raleigh, NC | pregmi3@ncsu.edu | Expected graduation: Summer 2027
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Professional Summary

I am a PhD candidate in Geospatial Analytics at NC State University, using lidar, UAS photogrammetry, satellite imagery, GIS, and Python/R to study Hurricane Helene's terrain, drainage, and forest impacts through research at NC State and Oak Ridge National Laboratory. I combine technical analysis with interdisciplinary collaboration, teaching, and clear scientific communication to support geohazard assessment and infrastructure resilience.

Technical Skills

Analysis: Multi-temporal lidar; Structure-from-Motion; terrain and drainage change; remote sensing; UAS mapping; geospatial machine learning. **Tools:** Python, R, GRASS GIS, ArcGIS Pro/Online, QGIS, Google Earth Engine, Pix4D, Agisoft Metashape, CloudCompare.

Relevant Experience

Graduate Research and Teaching Assistant | NC State University | Aug 2023–Present

- Integrate multi-temporal lidar, SfM, and surface-water modeling to assess Hurricane Helene's terrain and drainage impacts in the Chimney Rock–Lake Lure corridor; develop lidar–UAS terrain-modeling methods and contribute to open-source GIS and geospatial LLM research.
- Submitted the first dissertation-chapter manuscript on storm-driven terrain and drainage reorganization to *Natural Hazards*.

Research Intern | Oak Ridge National Laboratory | May–Aug 2026

- Analyze aerial and UAV lidar to assess forest resistance, delayed canopy loss, and early recovery following Hurricane Helene, complementing dissertation research on terrain and drainage change.

Summer Research Assistant | NC State University ITRE | June–Aug 2024

- Integrated parachute systems on Mavic drones to support operations-over-people waiver work.

Graduate Research Assistant | Texas A&M University–Corpus Christi | Sept 2021–Aug 2023

- Conducted UAS flights and processed imagery; developed 2D/3D web maps and dashboards in ArcGIS Online and maintained a historical campus UAS imagery site.

Geomatics Engineer | NEA Engineering Company Ltd., Nepal | Jan–July 2021

- Used GIS and satellite analysis to evaluate transmission-line routes against distance, cost, and community-benefit criteria.

Education

- PhD, Geospatial Analytics | North Carolina State University | Expected Summer 2027
- MS, Geospatial Systems Engineering | Texas A&M University–Corpus Christi | 2023
- BE, Geomatics Engineering | Kathmandu University, Nepal | 2018

Scientific Communication and Outreach

- Hosted two-hour CSDMS clinics on GRASS GIS and Tangible Landscape (2025) and coastal evolution and inundation modeling in Jupyter (2024).
- Presented Helene research at IALE 2026 and AGU 2025, UAS visibility analysis at AGU 2024, and geospatial LLM research at IALE 2025.
- Led satellite-imagery activities for K–12 students with disabilities (2024–2025) and taught data science and machine learning to high-school students in Nepal (2020–2021).

Leadership and Recognition

- Advanced Track Winner, NC PSI Hackathon (2025): used climate data and AI/ML models to identify five crops suited to North Carolina; participated in crop/weed classification at the 2024 hackathon.
- President, Nepali Student Association, NC State (2026–Present); Vice President (2025–2026). External Affairs Officer, Geospatial Graduate Student Organization (2024–2025). AGU H3S Subcommittee member (2025–2026).
- Women in GIS Award (2025) | NC State University Graduate Fellowship (2023) | Best Paper Presented, UAS in Geomatics Conference, IIT Roorkee (2019)

Certifications

- Leadership Learning Institute, Graduate School, NC State University - March 2026
- Thriving in Interdisciplinary Teams, four-week workshop, Shelton Leadership Institute, NC State University - March 2026
- FAA Part 107 sUAS Remote Pilot (2024) | AUVSI Trusted Operator (2025)

Selected Publications

- **Regmi, P.,** White, CT., & Mitasova, H. (2026) Assessment of storm-driven terrain and drainage re-organization in a forested mountain landscape. (In review: Springer Nature, Natural Hazards)
- White, CT., **Regmi,P**., Reckling, W., & Mitasova, H. (2026) Volumetric Change Detection with SfM Photogrammetry from Rapid-Response Aerial Imagery after Hurricane Helene (In review: MDPI, Remote Sensing)
- **Regmi, P**., Starek, M. J., & Berryhill, J. (2023). Overview of the Uncrewed Aircraft System (UAS) Campus Survey Project at Texas A&M University-Corpus Christi. FIG Working Week 2023.
- **Regmi, P**., & Wagle, N. (2021). Encroachment of Forest Cover in Chitwan and Parsha Districts of Nepal: A Spatiotemporal Quantification Over 25 Years. Journal of Land Management and Geomatics Education, 3, 7–12.
- Awasthi, B., Karki, S., **Regmi, P**., Dhami, D. S., Thapa, S., & Panday, U. S. (2020). Analyzing the Effect of Distribution Pattern and Number of GCPs on Overall Accuracy of UAV Photogrammetric Results. In Proceedings of UASG 2019: Unmanned Aerial System in Geomatics, 1, 339–354.